**CERTIFICATE OF VERIFICATION**

**Note:** This table completed by ECC Registry.

| | |
|---|---|
| Project Name: | Enforcement Agency: |
| Dwelling Address: | Permit Number: |
| City and Zip Code: | Permit Application Date: |

**A. System Information**

ECC Rater to field-verify all system information, discrepancies to be noted by overwriting entry.

| | | |
|---|---|---|
| 01 | Space Conditioning System Identification or Name | |
| 02 | Space Conditioning System Location or Area Served | |
| 03 | Condenser (or package unit) Make or Brand | |
| 04 | Condenser (or package unit) Model Number | |
| 05 | Nominal Cooling Capacity (tons) of Condenser | |
| 06 | Condenser (or package unit) Serial Number | |
| 07 | Refrigerant Type | |
| 08 | Other Refrigerant Type (if applicable) | |
| 09 | Liquid Line Filter Drier Installed According to Manufacturer's Specifications (if applicable) | |
| 10 | System Installation Type | |
| 11 | Fault Indicator Display (FID) Status (Note: Even systems with a FID must have refrigerant charge verified by installer) | |
| 12 | Is the system of a type that the minimum airflow can be verified for all indoor units using an approved measurement procedure (RA3.3 or RA3.3.3)? | |
| 13 | Is the system of a type that approved refrigerant charge verification procedures can be used to verify compliance with the refrigerant charge verification requirements when temperatures are ≥ 55°F (RA3.2.2, or RA1)? | |
| 14 | Date of Refrigerant Charge Verification for this System | |
| 15 | Refrigerant Charge Verification Method Used | |
| 16 | Person Who Performed the Refrigerant Charge Verification Reported on this Certificate of Installation | |
| 17 | ECC Verification Compliance Requirement Status | |
| 18 | Refrigerant Charge Verification Method Used by ECC Rater | |

### B1. Metering Device Verification

ECC Rater is required to visually field verify all information from CF2R. Superheat Method can only be used on systems that do not have a variable metering device.

| | | |
|---|---|---|
| 01 | Refrigerant Metering Device | |
| 02 | Superheat Method Applicability Status | |

### B2. Metering Device Verification

ECC Rater is required to visually field verify all information from CF2R. Subcooling Method can only be used on systems that have a variable metering device.

| | | |
|---|---|---|
| 01 | Refrigerant Metering Device | |
| 02 | Subcooling Method Applicability Status | |

### C. Instrument Calibration

ECC Raters are required to calibrate their diagnostic tools. Procedures for instrument calibration are given in Reference Residential Appendix RA3.2.2 and RA3.2.2.2

| | | |
|---|---|---|
| 01 | Date of Digital Refrigerant Gauge Calibration | |
| 02 | Date of Digital Thermocouple Calibration | |
| 03 | Digital Refrigerant Gauge Calibration Status | |
| 04 | Digital Thermocouple Calibration Status | |

### D. Measurement Access Hole (MAH) Verification

ECC Raters are required to visually field verify MAH. Procedures for installing MAH are specified in Reference Residential Appendix RA3.2.2.3.

| | | |
|---|---|---|
| 01 | Method Used to Demonstrate Compliance with the Measurement Access Hole (MAH) Requirement | |

### E. Minimum System Airflow Rate Verification

Procedures for verifying minimum system airflow are specified in Reference Residential Appendix RA3.3.3.

| 01 | 02 | 03 |
|---|---|---|
| Indoor Unit Name or Description of Area Served | Minimum Required System Airflow Rate (cfm) | System Airflow Rate Verification Status |
| | | |
| | | |
| 04 | Compliance Statement: | |
| Notes: | | |

**F1. Data Collection for Superheat Method**

ECC Rater must independently collect all data in this section. Procedures for determining Refrigerant Charge using the Standard Charge Verification Procedure are given in Reference Residential Appendix RA3.2.2 and RA3.2.2.2

| | | |
|---|---|---|
| 01 | Lowest Return Air Dry-bulb Temperature that Occurred During the Refrigerant Charge Verification Procedure (°F) | |
| 02 | Measured Condenser Air Entering Dry-bulb Temperature ($T_{condenser, db}$) (°F) | |
| 03 | Outdoor Temperature Qualification Status | |
| 04 | Measured Return (evaporator entering) Air Dry-bulb Temperature ($T_{return, db}$) (°F) | |
| 05 | Measured Return (evaporator entering) Air Wet-bulb Temperature ($T_{return, wb}$) (°F) | |
| 06 | Measured Suction Line Temperature ($T_{suction}$) (°F) | |
| 07 | Measured Suction Line Pressure ($P_{suction}$ - psig) | |
| 08 | Evaporator Saturation Temperature ($T_{evaporator, sat}$) from Digital Gauge or P-T Table using Line F07 (°F) | |
| 09 | Measured Superheat (Line F06 – Line F08) (°F) | |
| 10 | Target Superheat (from Table RA3.2-2, using F02 and F05) (°F) | |
| 11 | Compliance Statement: | |

**F2. Data Collection and Calculations for Subcooling Method**

ECC Rater must independently collect all data in this section. Procedures for Refrigerant Charge using the Standard Charge Verification Procedure are given in Reference Residential Appendix RA3.2.2.

| | | |
|---|---|---|
| 01 | Lowest Return Air Dry-bulb Temperature that Occurred During the Refrigerant Charge Verification Procedure (°F) | |
| 02 | Measured Condenser Air Entering Dry-bulb Temperature ($T_{condenser, db}$) | |
| 03 | Outdoor Temperature Qualification Status | |
| 04 | Measured Liquid Line Temperature ($T_{liquid}$) (°F) | |
| 05 | Measured Liquid Line Pressure ($P_{liquid}$) (psig) | |
| 06 | Condenser Saturation Temperature ($T_{condensor, sat}$) from Digital Gauge or P-T Table using Line F05 (°F) | |
| 07 | Measured Subcooling (Line F06 – Line F04) (°F) | |
| 08 | Target Subcooling from Manufacturer (°F) | |
| 09 | Compliance Statement: | |

## G. Metering Device Verification for Subcooling Method

ECC Rater must independently collect all data in this section. Procedures for the verification of proper metering device operation are specified in RA3.2.2.6.2

| | | |
|---|---|---|
| 01 | Measured Suction Line Temperature ($T_{suction}$) (°F) | |
| 02 | Measured Suction Line Pressure ($P_{suction}$) (psig) | |
| 03 | Evaporator Saturation Temperature ($T_{evaporator, sat}$) from Digital Gauge or P-T Table using line G02 (°F) | |
| 04 | Measured Superheat (Line G01 – Line G03) (°F) | |
| 05 | Measured Superheat (Line G04) is between 4°F and 25°F (inclusive) | |
| 06 | Measured Superheat (Line G04) is within Manufacturer's Specifications (if known) | |
| 07 | Compliance Statement: | |

## H. Weigh In Charge Procedure

ECC Rater Must Observe and Confirm All Data Collected. Procedures for Refrigerant Charge using the Weigh-in Charging Procedure are given in Reference Residential Appendix RA3.2.2.2 and RA3.2.3.

**The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.**

| | | |
|---|---|---|
| 01 | Measured Condenser Air Entering Dry-bulb Temperature ($T_{condenser, db}$) (°F) | |
| 02 | Specify the Method of Weigh-in | |
| 03 | Manufacturer's Standard Charge for Condenser (lbs, oz.) | |
| 04 | Manufacturer's Standard Liquid Line Length (ft) | |
| 05 | Manufacturer's Standard Liquid Line Diameter (in) | |
| 06 | Manufacturer's Standard Indoor Coil Size (tons) | |
| 07 | Installed Liquid Line Length (ft) | |
| 08 | Installed Liquid Line Diameter (in) | |
| 09 | Installed Indoor Coil Size (tons) | |
| 10 | Charge Adjustment to Standard Charge from Manufacturer's Specifications (ounces, positive = add, negative = remove) | |
| 11 | Refrigerant Required to be Weighed in by the Installer (lbs, oz) | |
| 12 | Refrigerant Weighed in by Installer (lbs, oz) | |
| 13 | Verification Status: (Note: If Verification Status for this table indicates "Fail", the reason shall be described in the correction notes for this table.) | |
| Correction Notes: | | |

## I. Weigh In Charge Procedure – Additional Requirements

**The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.**

| | | |
|---|---|---|
| 01 | If refrigerant line connections require welding, the system is braised with dry nitrogen in the lines and indoor coil. | |
| 02 | i. In all cases where the OEM instructions call for checking for gas leaks with vacuum, prior to introducing refrigerant, system is evacuated to 500 microns or less and, when isolated, has risen no more than 300 microns after 5 minutes.<br>ii. In all cases where the OEM instructions call for checking for gas leaks with nitrogen gas, the system was pressurized to the manufacturer's specified pressure and if the pressure could not be maintained, leaks were located and fixed. | |
| 03 | Observation and documentation of the vacuum and pressurization tests are not required if no fittings (other than the fitting to the compressor) are compression or flare fittings. | |
| 04 | The calculated weight adjustment for lineset length is based on the length and diameter of the lineset. | |
| 05 | The calculated weight adjustment for coil size is based on manufacturer instructions. | |
| 06 | The actual total weight adjustment is equal to the sum of the calculated weight adjustments for lineset and coil size. | |
| 07 | The calculated and actual total weights of refrigerant in the system are recorded on or near the nameplate label, in indelible ink or other permanent means. | |
| 08 | Verification Status:<br>(Note: If Verification Status for this table indicates "Fail", the reason shall be described in the correction notes for this table.) | |
| Correction Notes: | | |

## J. Determination of ECC Verification Compliance

All applicable sections of this document shall indicate compliance with the specified verification protocol requirements in order for this Certificate of Verification as a whole to be determined to be in compliance.

| | |
|---|---|
| 01 | |

*SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS*



## DOCUMENTATION AUTHOR'S DECLARATION STATEMENT

1. I certify that this Certificate of Verification documentation is accurate and complete.

| | |
|---|---|
| Documentation Author Name: | Documentation Author Signature: |
| Company: | Date Signed: |
| Address: | CEA/ECC Certification Information (if applicable): |
| City/State/Zip: | Phone: |

## RESPONSIBLE PERSON'S DECLARATION STATEMENT

I certify the following under penalty of perjury, under the laws of the State of California:

1. The information provided on this Certificate of Verification is true and correct.
2. I am the certified ECC Rater who performed the verification identified and reported on this Certificate of Verification (responsible rater).
3. The installed features, materials, components, manufactured devices, or system performance diagnostic results that require Field verification identified on this Certificate of Verification comply with the applicable requirements in Reference Appendices RA2, RA3, and the requirements specified on the Certificate of Compliance for the building approved by the enforcement agency.
4. The information reported on applicable sections of the Certificate(s) of Installation (CF2R) signed and submitted by the person(s) responsible for the construction or installation conforms to the requirements specified on the Certificate(s) of Compliance (CF1R) approved by the enforcement agency.
5. I understand that a registered copy of this Certificate of Verification shall be posted, or made available with the building permit(s) issued for the building, and made available to the enforcement agency for all applicable inspections, and I will take the necessary steps to ensure this requirement is accomplished.
6. I understand that a registered copy of this Certificate of Verification is required to be included with the documentation the builder provides to the building owner at occupancy, and I will take the necessary steps to ensure this requirement is accomplished.

## BUILDER OR INSTALLER INFORMATION AS SHOWN ON THE CERTIFICATE OF INSTALLATION

| | |
|---|---|
| Company Name (Installing Subcontractor, General Contractor, or Builder/Owner): | |
| Responsible Builder or Installer Name: | CSLB License: |

## ECC PROVIDER DATA REGISTRY INFORMATION

| | |
|---|---|
| Sample Group Number (if applicable): | Dwelling Test Status in Sample Group (if applicable): |

## ECC RATER INFORMATION

| | |
|---|---|
| ECC Rater Company Name: | |
| Responsible Rater Name: | Responsible Rater Signature: |
| Responsible Rater Certification Number w/ this ECC Provider: | Date Signed: |

**CF2R-MCH-25-H User Instructions**

**Section A. System Information**

1. This information is automatically pulled from the Certificate of Installation (CF2R-MCH-25). If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail.
2. This information is automatically pulled from the Certificate of Installation (CF2R-MCH-25) If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail.
3. This information is automatically pulled from the Certificate of Installation (CF2R-MCH-25). If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail.
4. This information is automatically pulled from the Certificate of Installation (CF2R-MCH-25) If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail.
5. This information is automatically pulled from the Certificate of Installation (CF2R-MCH-25). If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail.
6. This information is automatically pulled from the Certificate of Installation (CF2R-MCH-25) If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail.
7. This information is automatically pulled from the Certificate of Installation (CF2R-MCH-25) If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail. Choose the type of refrigerant used by the system being verified. R-454, R-22 and R-410A are the most common, but other types may occasionally be encountered.
8. This information is automatically pulled from the Certificate of Installation (CF2R-MCH-25). If “Other” is chosen in A07, then installer will indicate the type of refrigerant being used. If R-454, R-22 or R-410A is being used (regardless of trade name, Puron, Genetron, etc.) it should be indicated in A07, not here. This row is only for refrigerants other than R-22 and R-410a. Documentation of other refrigerants should be requested. If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail.
9. This information is automatically pulled from the Certificate of Installation (CF2R-MCH-25). If applicable, a liquid line filter drier shall be installed according to the manufacturer’s specifications.
10. This information is automatically pulled from the Certificate of Installation (CF2R-MCH-25). These are defined in detail the Residential Compliance Manual. If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail. Indicate whether the HVAC system is Completely New, Replacement or an Alteration.
11. N/A.
12. This information is automatically pulled from the Certificate of Installation (CF2R-MCH-25). Most ducted split systems and package systems are of the type that minimum airflow can be verified using an approved measurement procedure. Examples of systems that do not meet this description are ductless systems. Selecting “No” here may subject the project to additional scrutiny by enforcement personnel.
13. This information is automatically pulled from the Certificate of Installation (CF2R-MCH-25) Most ducted split systems and package systems are of the type that approved refrigerant charge verification procedures detailed in Residential Appendix RA3.2.2 or RA1 can be used (i.e., Standard Charge Verification procedures). Examples of systems that may not meet this description are “mini splits” or variable refrigerant flow systems that may only be charged using weigh-in procedures. Selecting “No” here may subject the project to additional scrutiny.
14. ECC rater to input date of their refrigerant charge verification.
15. This information is automatically pulled from the Certificate of Installation (CF2R-MCH-25). The installer is to have selected the refrigerant charge verification method used from the choices provided:

- Superheat (outdoor temperature must be ≥ 55°F); this verification method can only be used when the outdoor temperature is at or above 55°F. It is only used on systems with fixed orifice refrigerant metering devices (non-variable metering devices). This method is detailed in Reference Appendix RA3.2.2.6.1. Systems verified using this method may be eligible for ECC verification compliance using Group Sampling
- Subcooling (outdoor temperature must be ≥ 55°F); this verification method can only be used when the outdoor temperature is at or above 55°F. It is only used on systems with variable metering devices (TXV or EXV). This method is detailed in Reference Appendix RA3.2.2.6.2. Systems verified using this method may be eligible for ECC verification compliance using Group Sampling.
- Weigh-in; this verification method can be used by the installer at any outdoor temperature allowed by the equipment manufacturer. This method is detailed in Reference Appendix RA3.2.3. Systems verified using this method are NOT eligible for ECC verification compliance using Group Sampling.
- New Package Unit Factory Charge; the installer should choose this option when a new package unit is being installed that has an AHRI rating. This helps ensure that the unit was properly charged at the factory. ECC verification of refrigerant charge may not be required in this case.

16. This information is automatically pulled from the Certificate of Installation (CF2R-MCH-25). The installer (or rater) is to have identified who performed the verification that is documented on the Certificate of Installation. Note that ECC verification compliance by Group Sampling requires that the installer perform their own refrigerant charge verification as part of the installation of the equipment prior to the system being put into a sample group for possible selection by a ECC rater for verification. If Group Sampling is not intended, the ECC Rater may perform the refrigerant charge verification on behalf of the Installing Contractor (applies to any method but Weigh-In) and the Rater will enter same results on both the CF2R and CF3R.
17. This information is automatically pulled from the Certificate of Installation (CF2R-MCH-25). The Group Sampling status is automatically displayed based on the input results of A15 and A16 on the CF2R. Group Sampling procedures are detailed in Residential Appendix RA2.6.3.
18. Specify the refrigerant charge verification used by the ECC rater. Choices vary depending on what method was specified in Row A11, A12, and A15.

**Section B1 and B2. Metering Device Verification**

1. This information is automatically pulled from the Certificate of Installation (CF2R-MCH-25). Installer is to have selected the correct metering device used on the system being verified. This will check against the refrigerant charge verification method selected in A15. An error message will appear in B02 if the wrong verification method may have been selected. Superheat verification can only be used on systems with fixed orifice and Subcool verification can only be used on systems with variable metering devices (TXV or EXV). This entry must match installed system to pass.
2. This information is automatically pulled from the Certificate of Installation (CF2R-MCH-25). Superheat verification can only be used on systems with fixed orifice and Subcool verification can only be used on systems with variable metering devices (TXV or EXV).

**Section C. Instrument Calibration**

1. Enter the date of most recent Digital Refrigerant Gauge Calibration Field Check. Analog gauges are not allowed for verification purposes under the 2025 Standards. Specification for pressure gauges is found in Residential Appendix RA3.2.2.2.3. Procedures for the field check procedure are detailed in RA3.2.2.4.2. Calibration field check must happen at least once every 30 days.
2. Enter the date of the most recent Digital Thermocouple Calibration. Specifications for thermocouples and temperature sensors can be found in Residential Appendix RA3.2.2.2.2. Procedures for calibration are detailed in RA3.2.2.4.1. Calibration must happen at least once every 30 days.
3. Digital Refrigerant Gauge Calibration status will appear automatically. If the date entered in C01 is more than 30 days prior to date of verification this row will indicate that calibration is required and you will not be allowed to continue filling out this document until calibration is performed.
4. Digital Thermocouple Calibration status will appear automatically. If the date entered in C02 is more than 30 days prior to date of verification this row will indicate that calibration is required and you will not be allowed to continue filling out this document until calibration is performed.

**Section D. Measurement Access Hole (MAH) Verification**

1. This information is automatically pulled from the Certificate of Installation (CF2R-MCH-25). Installer is to have indicated the method used to demonstrate compliance with the MAH requirement by selecting the appropriate method from the drop down list. Procedures for installing MAH's are detailed in RA3.2.2.3. Selecting that the MAH cannot be installed consistent with Figure 3.2-1 may result in additional scrutiny by enforcement personnel.) If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail. For Weigh-in verification methods only If A12 = NO, then system is exempt from the MAH requirement and a special message will show up here.

**Section E. Minimum System Airflow Rate Verification**

1. This information is automatically calculated based on the information given in A10. This is the target minimum system airflow required for the system being verified.
2. This information is automatically calculated based on the MCH-23 or MCH-28, which documents the measured airflow (or alternative method) of the system being verified. If the measured airflow is not adequate it will not comply with the airflow requirements and refrigerant charge verification cannot be performed. For Weigh-in verification methods only If A12 = NO, then system is exempt from the airflow rate requirement and a special message will show up here.

**Section F1. Superheat Charge Verification Method – Data Collection**

1. The Rater must independently collect this data. Measure and record the lowest return air dry-bulb temperature that occurred during the refrigerant charge procedure in °F. This temperature must remain above 70°F during the verification procedure. This requirement is detailed in Residential Appendix RA3.2.2.5.
2. The Rater must independently collect this data. Measure and record the condenser air dry-bulb temperature ($T_{condenser}$) in °F. This value is used to determine the target superheat from table RA3.2-2. This value must be at least 55°F and no more than 115°F to use the Superheat Charge Verification Method.
3. If a value less than 55°F or greater than 115°F is entered in F02 the Superheat Method cannot be used.
4. The Rater must independently collect this data. Measure and record the return air dry-bulb temperature ($T_{return,db}$) in °F. This measurement is taken at the MAH (or alternate location specified in F01. This procedure is detailed in RA3.2.2.5.

5. The Rater must independently collect this data. Measure and record the return air wet-bulb temperature ($T_{return,wb}$) in ° F. This measurement is taken at the MAH (or alternate location specified in F01. This procedure is detailed in RA3.2.2.5. This value is used to determine the target superheat from table RA3.2-2.
6. The Rater must independently collect this data. Measure and record the suction line temperature ($T_{suction}$) in °F. This procedure is detailed in RA3.2.2.5. This value is used to calculate the measured superheat.
7. The Rater must independently report this data. This procedure is detailed in RA3.2.2.5. This value is used to determine the evaporator saturation temperature ($T_{evaporator,sat}$) from a pressure temperature chart for the appropriate refrigerant (can be internal to a digital gauge), which is entered into F08.
8. The Rater must independently collect this data. Enter the evaporator saturation temperature ($T_{evaporator,sat}$) from the digital gauge or a separate pressure-temperature chart that corresponds to the suction line pressure entered in F07, in °F.
9. Measured superheat is automatically calculated as the difference between the suction line temperature (F06) and the evaporator saturation temperature (F08)
10. The Rater must independently report this data. Enter target superheat from Table RA3.2-2. This table requires values for the condenser air dry bulb temperature (F02) and the return air wet bulb temperature (F05)
11. System passes superheat method when F10 is within plus or minus 8°F of F09. Note that the target for the installer, on the CF2R-MCH-25 is plus or minus 5°F.

**Section F2. Subcooling Charge Verification Method – Data Collection**

1. The Rater must independently collect this data. Measure and record the lowest return air dry-bulb temperature that occurred during the refrigerant charge procedure in °F. This temperature must remain above 70°F during the verification procedure. This requirement is detailed in Residential Appendix RA3.2.2.5.
2. The Rater must independently collect this data. Measure and record the condenser air dry-bulb temperature ($T_{condenser}$) in °F. This value must be at least 55°gF and no more than 115°F to use the Subcooling Charge Verification Method.
3. If a value less than 55°F or greater than 115°F is entered in F02 the Subcooling Method cannot be used.
4. The Rater must independently collect this data. Measure and record the liquid line temperature ($T_{liquid}$) in °F. This procedure is detailed in RA3.2.2.5. This value is used to calculate the measured subcool temperature.
5. The Rater must independently collect this data. Measure and record the liquid line pressure ($P_{liquid}$) in psig. This procedure is detailed in RA3.2.2.5. This value is used to determine the condenser saturation temperature ($T_{condenser,sat}$) from a pressure temperature chart for the appropriate refrigerant (can be internal to a digital gauge), which is entered into F06.
6. Enter the condenser saturation temperature ($T_{condenser,sat}$) from the digital gauge or a separate pressure-temperature chart that corresponds to the liquid line pressure entered in F05, in °F.
7. Measured Subcooling is automatically calculated as the difference between the liquid line temperature (F04) and the condenser saturation temperature (F06)
8. The Rater must independently collect this data. Enter target subcooling from manufacturer. This may be a challenge to find for older equipment. Internet searches can sometimes result in archived equipment specifications for the equipment in question, or sometimes a very similar model. If the manufacturer's target cannot be found the Commission's Executive Director may provide additional guidance for compliance.

9. System passes Subcooling method when F08 is within plus or minus 6°F of F07. Note that the target for the installer, on the CF2R, is plus or minus 3°F.

**Section G. Metering Device Verification for Subcooling Method**

1. The Rater must independently collect this data. Measure and record the suction line temperature ($T_{suction}$) in °F. This procedure is detailed in RA3.2.2.5. This value is used to calculate the measured superheat.
2. The Rater must independently collect this data. Measure and record the suction line pressure ($P_{suction}$) in psig. This procedure is detailed in RA3.2.2.5. This value is used to determine the evaporator saturation temperature ($T_{evaporator,sat}$) from a pressure temperature chart for the appropriate refrigerant (can be internal to a digital gauge), which is entered into G03.
3. Enter the evaporator saturation temperature ($T_{evaporator,sat}$) from the digital gauge or a separate pressure-temperature chart that corresponds to the suction line pressure entered in G02, in °F.
4. Measured superheat is automatically calculated as the difference between the suction line temperature (G01) and the evaporator saturation temperature (G03).
5. There are two possible criteria for passing. If the manufacturer’s specification is known it should be used, otherwise the CEC requirement is that the superheat be between 3°F and 26°F, inclusive. This row checks the CEC requirement.
6. If the manufacturer’s target superheat for ensuring proper metering device operation is known, it supersedes the CEC requirement of being between 3°F and 26°F. If “Yes, documentation to be provided upon request.” is selected, the installer should be prepared to provide documentation for the target values used.
7. There are two possible criteria for passing. If the manufacturer’s specification is known it should be used, otherwise the CEC requirement is that the superheat be between 3°F and 26°F, inclusive. If “Yes, documentation to be provided upon request.” is selected in G06, the installer should be prepared to provide documentation for the target values used.

**Section H. Weigh In Charge Procedure**

1. ECC rater must visually observe the installer taking this measurement and confirm that correct values are entered into the CF2R. Measure and record the outside air dry-bulb temperature in °F. This will affect the procedures that may be used for ECC verification.
2. ECC rater must confirm that correct values are entered into the CF2R. Specify the method of weigh-in. There are two options that may be used. One is to add or remove a small, weighed portion of refrigerant from a factory charged unit (Charge Adjustment). The other is to weigh the entire charge of refrigerant before introducing it into the system (Total Charge). Select either one. Note: The amount of refrigerant in systems that are not newly installed cannot be assumed to be the factory charge. Altered systems using existing refrigerant must use the Total Charge method. Only new, factory installed equipment can utilize the Charge Adjustment method.
3. ECC rater must confirm that correct values are entered into the CF2R. Enter the Manufacturer’s Standard Charge for condenser in pounds and ounces. This is the amount of refrigerant that the manufacturer specifies for a “standard” installation (typical coil match, typical line set size and length). For the Charge Adjustment method, this is the amount of refrigerant that factory charges the system to. Rater should request to see manufacturer’s documentation to support this value.
4. ECC rater must confirm that correct values are entered into the CF2R. The Manufacturer’s Standard Charge, specified in H03 is based on a standard liquid line length, typically 25 feet. Enter the value here, in feet. Be prepared to provide manufacturer’s documentation to support this value.

5. ECC  rater must confirm that correct values are entered into the CF2R. The Manufacturer’s Standard Charge, specified in H03 is based on a standard liquid line diameter. Enter the value here, in inches (for example: 1/4”, 3/8”, etc.). Rater should request to see manufacturer’s documentation to support this value.
6. ECC rater must confirm that correct values are entered into the CF2R. The Manufacturer’s Standard Charge, specified in H03 is based on a standard indoor (evaporator) coil size. Enter the value here, in tons. Rater should request to see manufacturer’s documentation to support this value.
7. ECC rater must confirm that correct values are entered into the CF2R. Enter the length of the liquid line installed on the system being verified, in feet. This value must be compared to the standard liquid line length entered in H04 and used to determine if the Manufacturer’s Standard Charge entered in E03 is appropriate.
8. ECC rater must confirm that correct values are entered into the CF2R. Enter the diameter of the liquid line installed on the system being verified, in inches (for example: 1/4”, 3/8”, etc.). This value must be compared to the standard liquid line diameter entered in H05 and used to determine if the Manufacturer’s Standard Charge entered in H03 is appropriate.
9. ECC rater must confirm that correct values are entered into the CF2R. Enter the size of the indoor (evaporator) coil installed on the system being verified, in tons. This value must be compared to the standard coil size entered in H06 and used to determine if the Manufacturer’s Standard Charge entered in H03 is appropriate.
10. ECC rater must confirm that correct values are entered into the CF2R. Enter the Charge Adjustment to Standard Charge, in ounces. This is the amount of refrigerant that the manufacturer specifies to add to, or remove from, the Manufacturer’s Standard Charge entered in H03. This value must come from manufacturer’s specifications using the standard values entered in H04 through H06 to the installed values entered in H07 through H09. If refrigerant is to be added, this value should be a positive number. If refrigerant is to be removed, this value should be a negative number. Rater should request to see manufacturer’s documentation to support this value.
11. ECC rater must confirm that correct values are entered into the CF2R. This value is calculated automatically. If “Charge Adjustment” was specified in H02, then the value shown here will be the same as the value shown in H10. This is the amount of weighed refrigerant that will be added or removed from the factory charged unit. If refrigerant is to be added, this value should be a positive number.  If refrigerant is to be removed, this value should be a negative number. If “Total Charge” was specified in H02, then the value shown here will be the value in H03 added to the value in H10. This is the total amount of refrigerant that will be in the system, all of which must be weighed before introducing into the system.
12. ECC rater must confirm that correct values are entered into the CF2R. Enter the amount of refrigerant weighed and added to, or removed from, system. If refrigerant is to be added, this value should be a positive number. If refrigerant is to be removed from a factory charged system, this value should be a negative number. This value must match the value in H11 for the system to pass.
13. ECC rater to indicate whether system passes or not. If not, use the next line to provide notes as to why system did not pass.

**Section I. Weigh In Charge Verification – Additional Requirements**

1. The Rater must confirm refrigerant line connections were checked for welding as required by the requirement for brazing lines charged with dry nitrogen is specified in Residential Appendix RA3.2.3.1.5.

2. The Rater must confirm refrigerant lines for leaks with nitrogen gas by pressurized to the manufacturer's specified pressure and if the pressure cannot be maintained, leaks shall be located and fixed.
3. The Rater must confirm refrigerant lines were checked for leaks by evacuating to 500 microns or less and rising by no more than 300 microns after 5 minutes as required by Residential Appendix RA3.2.3.1.5.
4. The Rater must confirm that Observation and documentation of the vacuum and pressurization tests are not required if no fittings (other than the fitting to the compressor) are compression or flare fittings is specified in Residential Appendix RA3.2.3.1.5.
5. The Rater must confirm that the calculated weight adjustment for lineset length is based on the length and diameter of the lineset is specified in Residential Appendix RA3.2.3.1.5.
6. The Rater must confirm that the calculated weight adjustment for coil size is based on manufacturer instructions is specified in Residential Appendix RA3.2.3.1.5.
7. The Rater must confirm that the actual total weight adjustment is equal to the sum of the calculated weight adjustments for lineset and coil size is specified in Residential Appendix RA3.2.3.1.5.
8. The Rater must confirm that the calculated and actual total weights of refrigerant in the system are recorded on or near the nameplate label, in indelible ink or other permanent means is specified in Residential Appendix RA3.2.3.1.5.

**A. System Information**
<<Static Text - see the top>>

| | | |
|---|---|---|
| 01 | Space Conditioning System Identification or Name | <<auto filled text: referenced from CF2R. If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail.>> |
| 02 | Space Conditioning System Location or Area Served | <<auto filled text: referenced from CF2R. If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail.>> |
| 03 | Condenser (or package unit) Make or Brand | <<auto filled text: referenced from CF2R. If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail.>> |
| 04 | Condenser (or package unit) Model Number | <<auto filled text: referenced from CF2R. If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail.>> |
| 05 | Nominal Cooling Capacity (tons) of Condenser | <<auto filled text: referenced from CF2R. If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail.>> |
| 06 | Condenser (or package unit) Serial Number | <<auto filled text: referenced from CF2R. If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail.>> |
| 07 | Refrigerant Type | <<auto filled text: referenced from CF2R. Possible entries are “R-454B”, “R-22” or “R-410a” or "other". If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail.>> |
| 08 | Other Refrigerant Type (if applicable) | << if A07 value = R-454B, R-22 or R-410A then value in this field = N/A; elseif value in A07= other, then user input text in this field to identify the refrigerant type>> |
| 09 | Liquid Line Filter Drier Installed According to Manufacturer’s Specifications (if applicable) | <<auto filled text: referenced from CF2R. Possible entries are “Yes” or “NA”. If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail>> |
| 10 | System Installation Type | <<auto filled text: referenced from CF2R. Possible entries are “New”, “Replacement”, or “Alteration”. If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail.>> . |
| 11 | Fault Indicator Display (FID) Status<br>(Note: Even systems with a FID must have refrigerant charge verified by installer) | << N/A>> |
| 12 | Is the system of a type that the minimum airflow can be verified for all indoor units using an approved measurement procedure (RA3.3 or RA3.3.3)? | <<auto filled text: referenced from CF2R. Possible entries are “yes” or “no” If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail.>> |
| 13 | Is the system of a type that approved refrigerant charge verification procedures can be used to verify compliance with the refrigerant charge verification requirements when temperatures are ≥ 55°F (RA3.2.2, or RA1)? | <<auto filled text: referenced from CF2R. Possible entries are “yes” or “no” If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail>> |
| 14 | Date of Refrigerant Charge Verification for this System | <<user input: date: use validated date format>> |
| 15 | Refrigerant Charge Verification Method Used | <<auto filled text: referenced from CF2R. Possible entries are:<br>• Superheat (outdoor temperature must be ≥ 55 degF); or<br>• Subcooling (outdoor temperature must be ≥ 55 degF); or<br>• Weigh-in with Installer independent; or<br>• Weigh-in with ECC Rater observation; or<br>• New Package Unit Factory Charge >> |
| 16 | Person Who Performed the Refrigerant Charge Verification Reported on this Certificate of Installation | • <<auto filled text: referenced from CF2R. Possible entries: HVAC System Installer or ECC Rater.>> |

| | | |
|---|---|---|
| 17 | ECC Verification Compliance Requirement Status | <<auto filled text: referenced from CF2R. Possible entries:<br>"System does not qualify for Group Sampling"; or<br>"System qualifies for Group Sampling.">> |
| | Generate list for next row (this is hidden from user) | If A15 = "Superheat", then list =<br>Superheat<br><br>Else, If A15 = "Subcooling", then list =<br>Subcooling<br><br>Else, If A15 = "Weigh-In with Installer independent", then list =<br>Superheat<br>Subcooling<br><br>Else if A15 = "Weigh-in with ECC Rater observation", then list =<br>Weigh-In Observation<br><br>Else, If A15 = "New Package Unit Factory Charge", then do not proceed. A CF3R-MCH-25 is not required when a CF2R-MCH-25f is used.<br>Else, If A12 = "No", or A13 = "No", then list =<br>Weigh-In Observation |
| 18 | Refrigerant Charge Verification Method Used by ECC Rater | <<user pick one from list generated in previous row>>> |

**B1. Metering Device Verification**
<<Static Text - see the top>>
<<If A18 == Superheat this section is required Else only display Section title and this note: This section does not apply to this project.>>

| | | |
|---|---|---|
| 01 | Refrigerant Metering Device | <<auto filled text: referenced from CF2R. Possible entry:<br>• Fixed orifice. If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail>> |
| 02 | Superheat Method Applicability Status | << If B01 = Thermostatic Expansion Valve or Electronic Expansion Valve; then display text:<br>"Superheat Method is not applicable to this system" (do not proceed);<br>else, display text:<br>" Superheat Method is applicable to this system">> |

**B2. Metering Device Verification**
<<Static Text - see the top>>
<<If A18 == Subcooling this section is required Else only display Section title and this note: This section does not apply to this project.>>

| | | |
|---|---|---|
| 01 | Refrigerant Metering Device | <<autofill, reference from C2R. Choices are Thermostatic Expansion Valve or Electronic Expansion Valve. If installed system does not match this entry, it can be overwritten by rater but it will be flagged as a possible fail.>> |
| 02 | Subcooling Method Applicability Status | << If B01 = Thermostatic Expansion Valve or Electronic Expansion Valve; then display text:<br>"Subcooling Method is applicable to this system";<br>else, display text:<br>"Subcooling Method is not applicable to this system" (do not proceed)>> |

**C. Instrument Calibration**
<<Static Text - see the top>>

| | | |
|---|---|---|
| 01 | Date of Digital Refrigerant Gauge Calibration | <<user input: date of calibration: use validated date format>> |
| 02 | Date of Digital Thermocouple Calibration | <<user input: date of calibration: use validated date format>> |
| 03 | Digital Refrigerant Gauge Calibration Status | <<if A14 compared to C01 is greater than one month, then display text:<br>"Digital Refrigerant Gauge requires Calibration (do not proceed)";<br>elseif A14 compared to C01 is ≥ 0 and ≤ one month; then display text: "calibration is current">> |
| 04 | Digital Thermocouple Calibration Status | <<if A14 compared to C02 is greater than one month, then display text:<br>"Digital Thermocouple Gauge requires Calibration (do not proceed)"<br>elseif A14 compared to C01 is ≥ 0 and ≤ one month; then display text: "calibration is current">> |

**D. Measurement Access Hole (MAH) Verification**
<<Static Text - see the top>>

| | | |
|---|---|---|
| 01 | Method Used to Demonstrate Compliance with the Measurement Access Hole (MAH) Requirement | << If A18 = WeighInObservation And A12 = No then display result=**The airflow rate measurement procedures in RA3.3 or RA3.3.3 are not applicable to this system, therefore compliance shall use ECC Rater observation (RA3.2.3.2) of the installer's weigh-in charging procedure(RA3.2.3.1); and compliance with MAH installation shall not be required.**<br>Else reference value from CF2R as default; allow user to override the default and pick one from list:<br>• "MAH installed and labeled consistent with Figure 3.2-1"; or<br>• "Return side of system is located entirely within conditioned space such that an accurate return air dry-bulb temperature can be taken at the return grille"; or<br>• "MAH cannot be installed consistent with Figure 3.2-1. An alternative location has been provided and clearly labeled"<br>• MAH is not installed. System does not comply >> |

**E. Minimum System Airflow Rate Verification**

<<Static Text - see the top>>

<<require 1 row of data for each indoor unit listed in the "ECC Verification Requirements for Duct Systems" table on the MCH-01>

| 01 | 02 | 03 |
|---|---|---|
| Indoor Unit Name or Description of Area Served | Minimum Required System Airflow Rate (cfm) | System Airflow Rate Verification Status |
| <<reference value from the "ECC Verification Requirements for Duct Systems" table on the MCH-01 for the "SC System Description of Area Served" value in A02>> | << if A12 = no, then N/A;<br>Else if the CF2R-MCH-01 indicates a MCH-28 is required for alternate minimum airflow rate compliance, then text value in this field=Verification of Table 150.0-B or C Alternative Return Duct Design Criteria is Required;<br><br>Else reference numeric xxxx value from applicable MCH-23 field for the indoor unit in E01<br>Applicable MCH23 field found in Forced Air System Airflow Rate Measurement Section E1, E2 or E3 field 03 AirflowRequiredMinimumTarget | <<If if A12 = no and A18 = WeighinObservation result = **The airflow rate measurement procedures in RA3.3 or RA3.3.3 are not applicable to this system, therefore compliance shall use ECC Rater observation (RA3.2.3.2) of the installer's weigh-in charging procedure (RA3.2.3.1)**;<br><br>Else if the CF2R-MCH-01 indicates a MCH-28 is required for alternate minimum airflow rate compliance, then<br><br>if the system has a registered CF3R-MCH-28 that indicates compliance with Table 150.0-B or C return duct design requirements, then result =**system complies using Table 150.0-B or C alternative return duct design criteria**.<br><br>else result=**System does not comply. A registered CF3R-MCH-28 is required** (do not allow this MCH-25 to be registered).<br><br>elseif the CF2R-MCH-01 indicates a MCH-23 is required for minimum airflow rate compliance, then<br><br>if this system has a registered CF2R-MCH-23 that meets the compliance criterion in E02, then result = **System complies with minimum airflow rate requirements**;<br><br>elseif A10=Alteration, then<br><br>if the system complies with the alternative airflow compliance method on a registered CF2R-MCH23; then result =**system complies using the alternative remedial actions specified in RA3.3.3.1.5**. **This System does not qualify for Group Sampling.**<br><br>else result=**System does not comply. A registered CF3R-MCH-23 for this system is required** (do not allow this MCH-25 to be registered)>> |
| | | |
| 04 | Compliance Statement: << If all indoor units listed in E01 indicate a compliant result in E03, then text result= "SC system complies with Minimum System Airflow Rate Verification"; else text result= "SC system does not comply with with Minimum System Airflow Rate Verification", and do not allow this MCH-25 to be registered. | |

Notes:

| F1. Data Collection for Superheat Method<br><<Static Text - see the top>><br><<If A18 == Superheat this section is required Else only display Section title and this note: This section does not apply to this project.>> | | |
|---|---|---|
| 01 | Lowest Return Air Dry-bulb Temperature that Occurred During the Refrigerant Charge Verification Procedure (°F) | <<user input: numeric: xxx.x, (in order to have a verification that complies, the return air drybulb temperature must remain above 70F during the verification procedure), range = 0 to 130>> |
| 02 | Measured Condenser Air Entering Dry-bulb Temperature ($T_{condenser, db}$) (°F) | <user input: numeric: xxx.x, range = 0 to 130>> |
| 03 | Outdoor Temperature Qualification Status | <<if F02<55F or greater than 115F, then display text: "Superheat refrigerant charge verification method is not allowed to be used when the outdoor temperature is less than 55F or greater than 115F", do not proceed>> |
| 04 | Measured Return (evaporator entering) Air Dry-bulb Temperature ($T_{return, db}$) (°F) | <<user entry, check range = 0 to 130>> |
| 05 | Measured Return (evaporator entering) Air Wet-bulb Temperature ($T_{return, wb}$) (°F) | <<user entry, check range = 0 to 130>> |
| 06 | Measured Suction Line Temperature ($T_{suction}$) (°F) | <<user entry, check range = -40 to 150>> |
| 07 | Measured Suction Line Pressure ($P_{suction}$ - psig) | <<user entry, check range = 0 to 400>> |
| 08 | Evaporator Saturation Temperature ($T_{evaporator, sat}$) from Digital Gauge or P-T Table using Line F07 (°F) | <<user entry, check range = -40 to 150>> |
| 09 | Measured Superheat (Line F06 – Line F08) (°F) | <<temperature, calculated (F06 – F08)>> |
| 10 | Target Superheat (from Table RA3.2-2, using F02 and F05) (°F) | <<user entry, check range = 0 to 50>> |
| 11 | Compliance Statement: <<if F01 ≥ 70, and ABS(F09 – F10) ≤ 8, then display text: "System complies with Refrigerant Charge Verification requirement by use of the Superheat Method"; else display text: "System does not comply" | |

| F2. Data Collection and Calculations for Subcooling Method<br><<Static Text - see the top>><br><<If A18 == Subcooling this section is required Else only display Section title and this note: This section does not apply to this project.>> | | |
|---|---|---|
| 01 | Lowest Return Air Dry-bulb Temperature that Occurred During the Refrigerant Charge Verification Procedure (°F) | <<user input: numeric: xxx.x, (in order to have a verification that complies, the return air drybulb temperature must remain above 70F during the verification procedure), range = 0 to 130>> |
| 02 | Measured Condenser Air Entering Dry-bulb Temperature ($T_{condenser, db}$) | <user input: numeric: xxx.x, check range), range = 0 to 130> |
| 03 | Outdoor Temperature Qualification Status | <<if F02<55F, then display text: " Subcooling refrigerant charge verification methods are not allowed to be used when the outdoor temperature is less than 55F", do not proceed>> |
| 04 | Measured Liquid Line Temperature ($T_{liquid}$) (°F) | <user entry, check range = -40 to 150> |
| 05 | Measured Liquid Line Pressure ($P_{liquid}$) (psig) | <user entry, check range = 0 to 800> |
| 06 | Condenser Saturation Temperature ($T_{condensor, sat}$) from Digital Gauge or P-T Table using Line F05 (°F) | <user entry, check range = -40 to 150> |
| 07 | Measured Subcooling (Line F06 – Line F04) (°F) | <<temperature, calculated (F06 – F04)>> |
| 08 | Target Subcooling from Manufacturer (°F) | <user entry, check range = 0 to 50> |
| 09 | Compliance Statement: <<if F01 ≥ 70, and ABS(F07 – F08) ≤ 6 and F07 ≥2, then display text: "System Refrigerant Charge Passes by Subcooling Method – Must also pass Metering Device Verification, next section." | |

### G. Metering Device Verification for Subcooling Method

<<Static Text - see the top>>

<<If A18 == Subcooling this section is required Else only display Section title and this note: This section does not apply to this project.>>

| | | |
|---|---|---|
| 01 | Measured Suction Line Temperature ($T_{suction}$) (°F) | <user entry, check range = -40 to 150> |
| 02 | Measured Suction Line Pressure ($P_{suction}$) (psig) | <user entry, check range = 0 to 400> |
| 03 | Evaporator Saturation Temperature ($T_{evaporator, sat}$) from Digital Gauge or P-T Table using line G02 (°F) | <user entry, check range = -40 to 150> |
| 04 | Measured Superheat (Line G01 – Line G03) (°F) | <<temperature, calculated, Line G01 – Line G03>> |
| 05 | Measured Superheat (Line G04) is between 4°F and 25°F (inclusive) | <<if 3 ≤ G04 ≤ 26 then display text "Passes CEC requirement">> |
| 06 | Measured Superheat (Line G04) is within Manufacturer's Specifications (if known) | <<user entry, choose "Not known", "Yes, documentation to be provided upon request", or "No">> |
| 07 | Compliance Statement: << If G05 = "Passes CEC requirement" and G06 = "Not known", or G06 = "Yes, documentation to be provided upon request", then display text: "Metering Device Verification Passes" | |

### H. Weigh In Charge Procedure

<<Static Text - see the top>>

<<If A18 = WeighInObservation this section is required Else only display Section title and this note: This section does not apply to this project.>>

| | | |
|---|---|---|
| 01 | Measured Condenser Air Entering Dry-bulb Temperature ($T_{condenser, db}$) (°F) | <<reference from CF2R. Rater to have observed the collection of this data.>> |
| 02 | Specify the Method of Weigh-in | <<reference from CF2R. Rater to have observed the collection of this data.>> |
| 03 | Manufacturer's Standard Charge for Condenser (lbs, oz.) | <<reference from CF2R. Rater to have observed the collection of this data.>> |
| 04 | Manufacturer's Standard Liquid Line Length (ft) | <<reference from CF2R. Rater to have observed the collection of this data.>> |
| 05 | Manufacturer's Standard Liquid Line Diameter (in) | <<reference from CF2R. Rater to have observed the collection of this data.>> |
| 06 | Manufacturer's Standard Indoor Coil Size (tons) | <<reference from CF2R. Rater to have observed the collection of this data.>> |
| 07 | Installed Liquid Line Length (ft) | <<reference from CF2R. Rater to have observed the collection of this data.>> |
| 08 | Installed Liquid Line Diameter (in) | <<reference from CF2R. Rater to have observed the collection of this data.>> |
| 09 | Installed Indoor Coil Size (tons) | <<reference from CF2R. Rater to have observed the collection of this data.>> |
| 10 | Charge Adjustment to Standard Charge from Manufacturer's Specifications (ounces, positive = add, negative = remove) | <<reference from CF2R. Rater to have observed the collection of this data.>> |
| 11 | Refrigerant Required to be Weighed in by the Installer (lbs, oz) | <<reference from CF2R. Rater to have observed the collection of this data.>> |
| 12 | Refrigerant Weighed in by Installer (lbs, oz) | <<reference from CF2R. Rater to have observed the collection of this data.>> |
| 13 | Verification Status: (Note: If Verification Status for this table indicates "Fail", the reason shall be described in the correction notes for this table.) | <<user input, pull down list: System Complies; System does not Comply>> |
| Correction Notes: <<user input, text, 200 characters>> | | |

| I. Weigh In Procedure – Additional Requirements<br><<If A18 = WeighInObservation this section is required Else only display Section title and this note:<br>This section does not apply to this project.>> | | |
|---|---|---|
| 01 | If refrigerant line connections require welding, the system is braised with dry nitrogen in the lines and indoor coil. | |
| 02 | i. In all cases where the OEM instructions call for checking for gas leaks with vacuum, prior to introducing refrigerant, system is evacuated to 500 microns or less and, when isolated, has risen no more than 300 microns after 5 minutes.<br>ii. In all cases where the OEM instructions call for checking for gas leaks with nitrogen gas, the system was pressurized to the manufacturer's specified pressure and if the pressure could not be maintained, leaks were located and fixed. | |
| 03 | Observation and documentation of the vacuum and pressurization tests are not required if no fittings (other than the fitting to the compressor) are compression or flare fittings. | |
| 04 | The calculated weight adjustment for lineset length is based on the length and diameter of the lineset. | |
| 05 | The calculated weight adjustment for coil size is based on manufacturer instructions. | |
| 06 | The actual total weight adjustment is equal to the sum of the calculated weight adjustments for lineset and coil size. | |
| 07 | The calculated and actual total weights of refrigerant in the system are recorded on or near the nameplate label, in indelible ink or other permanent means. | |
| 08 | Verification Status:<br>(Note: If Verification Status for this table indicates "Fail", the reason shall be described in the correction notes for this table.) | <<user input, pull down list: "Pass", "Fail">> |
| Correction Notes: <<user input, text, 200 characters>> | | |
| **The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.** | | |

| J. Determination of ECCVerification Compliance<br><<Static Text - see the top>> | |
|---|---|
| 01 | <<if D01≠ System does not comply; and E04≠ System does not comply; and F1 F11 or F2 F09 ≠ System does not comply and If present H13≠ System does not comply and I08 ≠ Fail; then display: Complies: All specified verification protocol requirements on this document are met; else display: Does not comply: One or more specified verification protocol requirements on this document are not met >> |